

16-bit magnetic navigation sensor manual

Modbus communication protocol

The sensor model to which this manual applies: CCF-NS16-2M、CCF-NS16-4M



1. Product introduction:

Support serial port output;

Support baud rate setting and output rate setting;

Support output mode setting;

Support N-pole and S-pole one key switching;

Support one key setting of sensing distance;

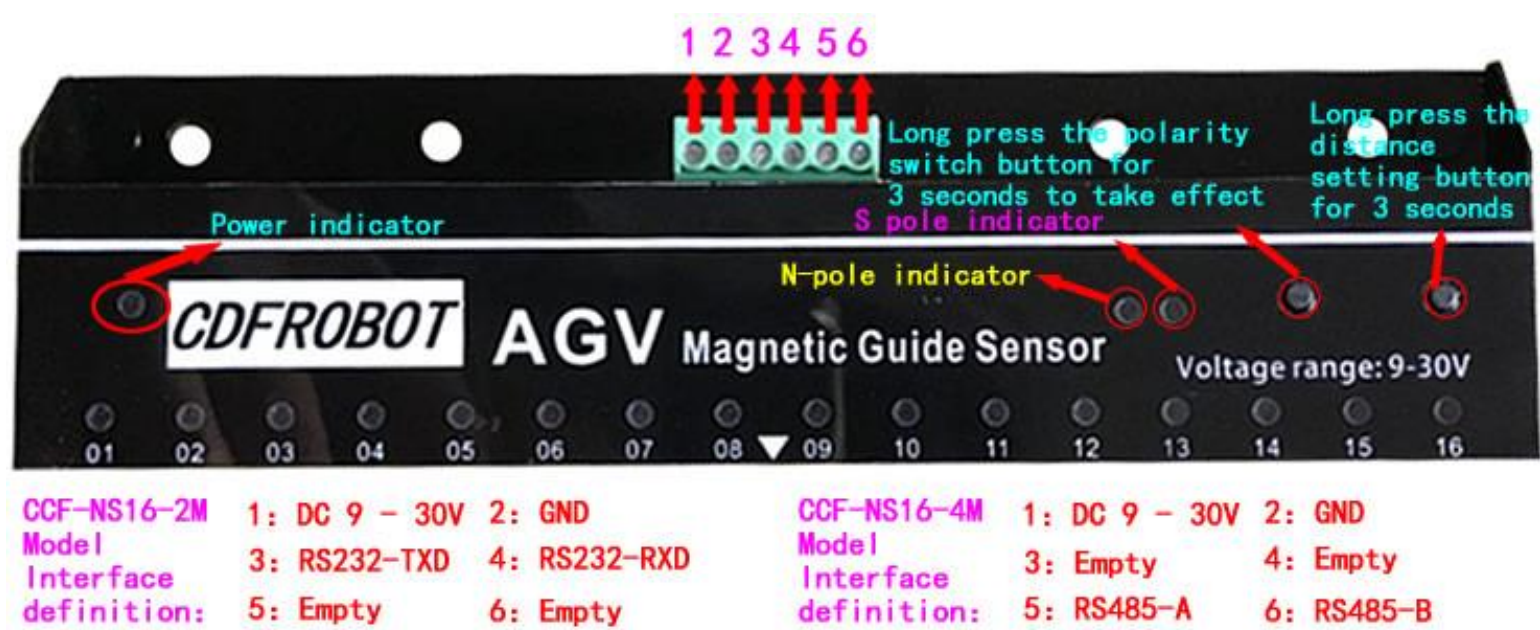
The standard 16 points are evenly distributed, and the distance between points is 10 mm;

- Signal status indicator display;
- Power status indicator display;
- Industrial three proofing design, wide voltage, low power consumption;
- High sensitivity semiconductor Hall sensor;

2. Electrical parameter description:

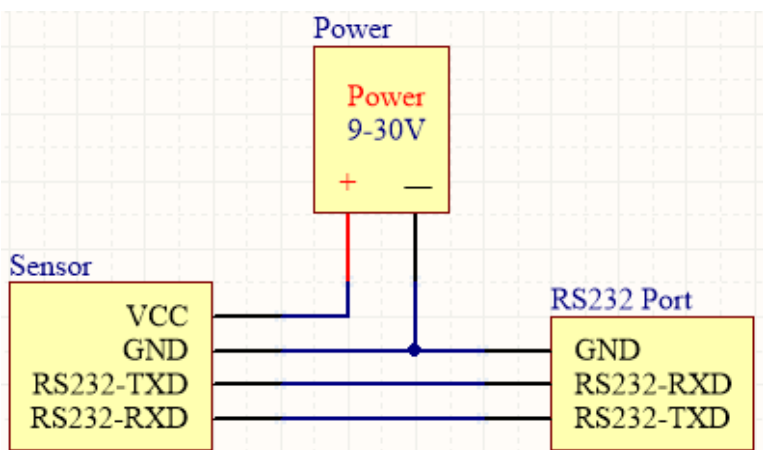
- Supply voltage: DC 9 - 30V;
- Maximum current consumption: 80mA;
- Output interface: RS232 or RS485 interface;
- Communication protocol: Modbus agreements;
- Detection effective distance: 5 - 85mm;
- Recommended installation height: 20 - 40mm;
- Response speed: 5ms;
- Accuracy: 5mm;
- Use temperature range: -40 ° C - +85 ° C;
- Working humidity: 10% - 90% relative humidity;

3. Interface definition:

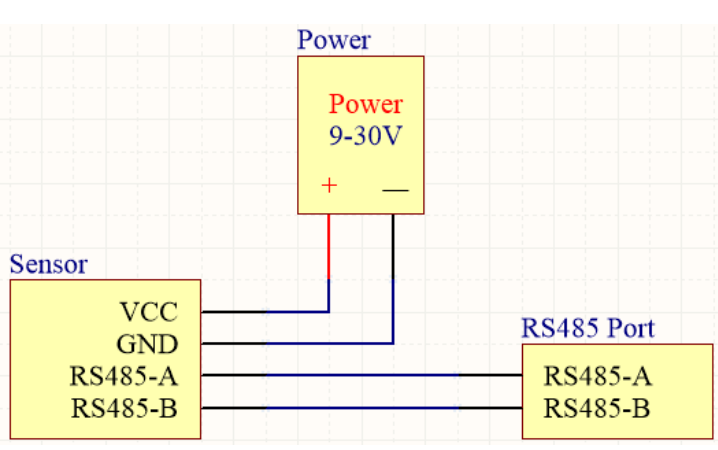


4. product electrical connection:

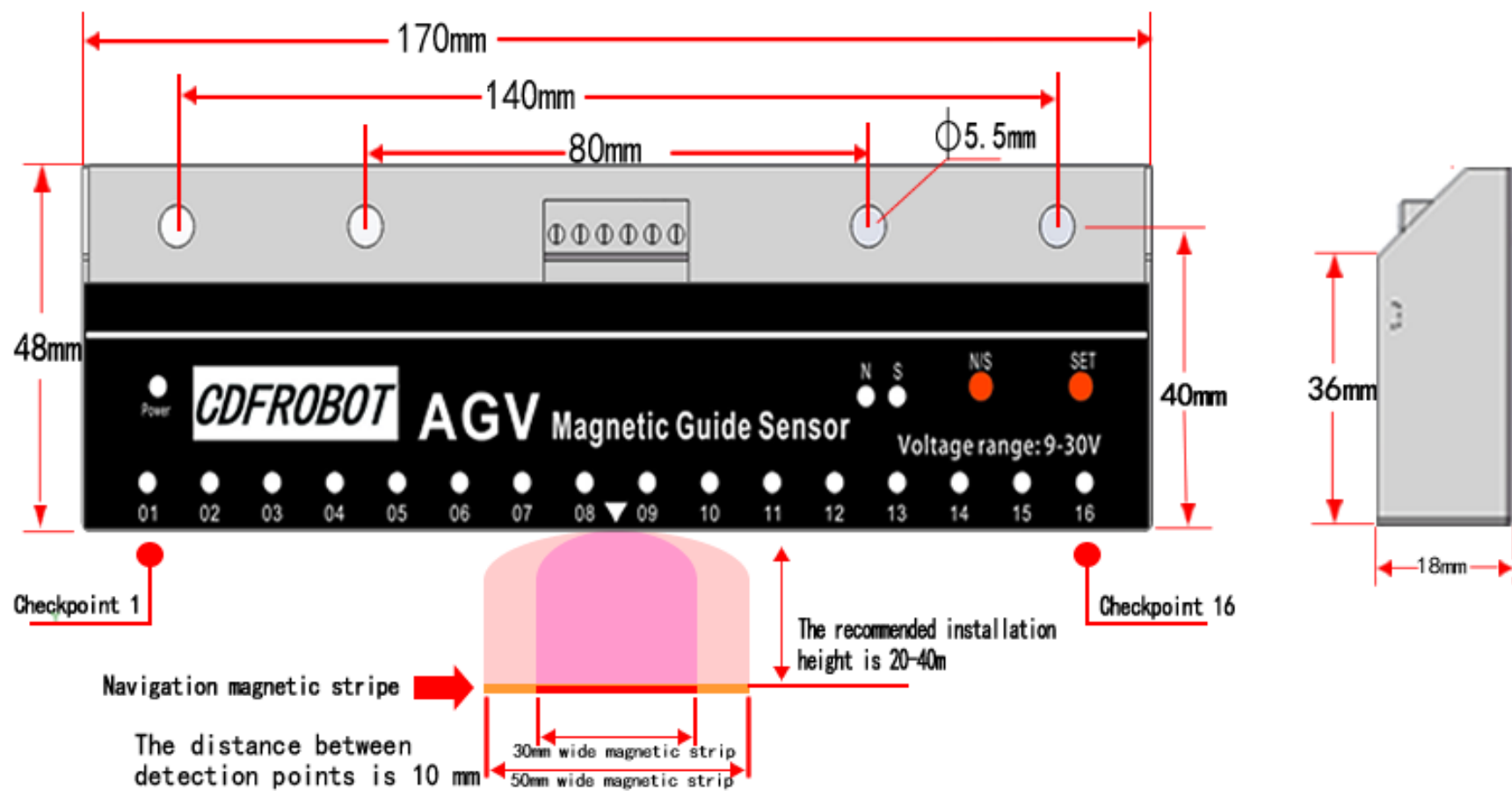
RS232 interface connection diagram:



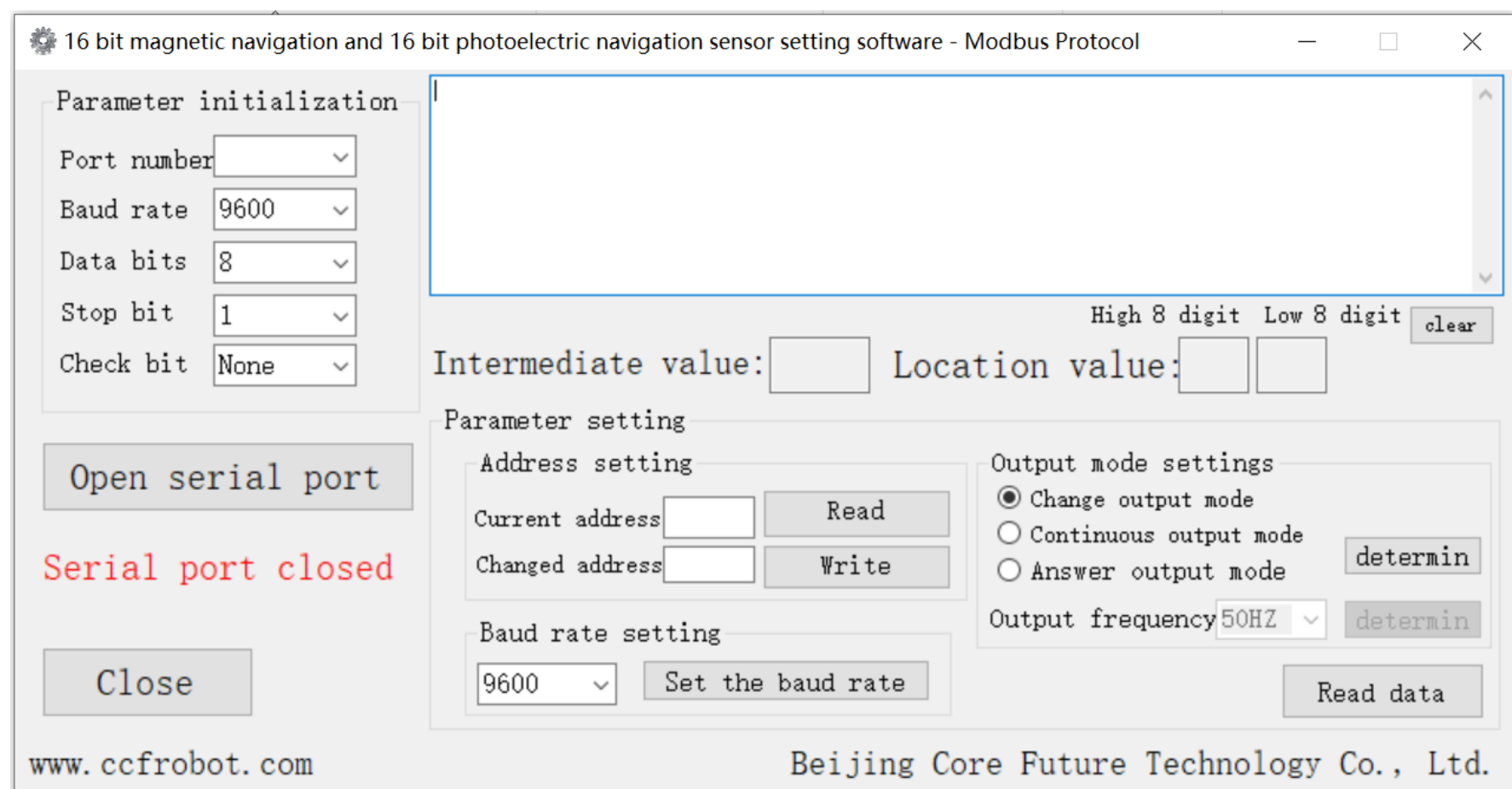
RS485 interface connection diagram:



5. Size description:



6. the upper computer settings software instructions:



- Parameter initialization: the default is shown in the figure;
- Open / close the serial port: select the correct com slogan to open the serial port normally, otherwise an error is reported; before plugging the USB serial port cable, close the serial port first;
- Intermediate value: display the value of the most intermediate point of the magnetic stripe; for example, when 3,4,5,6 points are sensed, the

intermediate value is 4.5; for example, when the sensing point is 3,4,5,6,7, the intermediate value is 5;

- Location value: displays the hexadecimal value of the current point sensed by the magnetic stripe;
- Current address: click the read address button to display the current address of the sensor (the physical address of the sensor is used as the slave address of the slave machine);
- Change address: input the address number to be set first, and then click write address to change the address of the sensor (the physical address of the sensor is used as the slave address of the slave machine);
- Set baud rate: select the baud rate to be set, and click the set baud rate button to change the baud rate of the sensor;
- Mode setting: select the corresponding output mode and click the OK button to take effect;
- Change output mode: that is, when the signal detected by the detection point of the sensor changes, it will output only when there is no change or output, and the latest detection data will be output immediately if there is a change (applicable to the situation that the processor uses serial port interrupt to receive data);
- Continuous output mode: the sensor outputs signal data continuously;
- Response output mode: Send a read command, and the sensor returns a detection data;
- Output frequency: the speed frequency of continuous output signal data of the sensor. Note: it is only valid in the continuous output mode;
- Read data: click the read data button to read the data content induced by the current sensor;
- **Note: after all the above functions are set successfully, the latest settings will be automatically saved after the sensor is powered on again;**

7. Communication Format and Communication Protocol Description:

1. Communication format: (8 data bits, 1 stop bit, no parity, default rate is 9600)

This product adopts standard Modbus RTU communication protocol, this product supports 03 06 function code, address: 1~254 can be set (factory default address is 1);

2. Host requests sensor data

For example: the host requests sensor data with device address 1

Host request: send data as 01 03 00 00 00 02 C4 0B								
01	03	00	00	00	02	C4	0B	
Device address	function code	Start register address high byte	Start register address low byte	High byte of read discrete data amount	The low byte of discrete data read	CRCL	CRCH	
Slave reply: Return data is 01 03 04 04 05 00 00 EB 02								
01	03	04	04	05	00	00	EB	02
Device address	function code	The number of bytes corresponding to the number of registers	Median value Integer bit	Median value Decimal places	High 8 bits Position value	Lower 8 bits Position value	CRCL	CRCH

3. address range 1 ~ 254 can be set (factory default address is 01) .Address modification need to re-power module to take effect. After the address is modified, new addresses are required for communication;

For example: write to change address 1 to address 2

Host request: Send data to 01 06 00 02 00 02 A9 CB （Change address 1 to 2）								
01	06	00	02	00	02	A9	CB	
Device address	function code	Preset register address high byte	Preset register address lower byte	Place data high byte	Place data low byte	CRCL	CRCH	
Slave reply: Return data is 01 06 00 02 00 02 A9 CB								

01	06	00	02	00	02	A9	CB
Device address	function code	Preset register address high byte	Preset register address lower byte	Place data high byte	Place data low byte	CRCL	CRCH

4. When modifying the baud rate, the baud rate of the current module should be used for communication. After the baud rate is modified, the module must be powered on again to take effect. Currently supports 1200,2400,4800,9600,19200,38400,57600,115200 baud rate, the module factory default baud rate is 9600; 2400 corresponds to 0x0001; 4800 corresponds to 0x0002; 9600 corresponds to 0x0003; 19200 corresponds to 0x0004; 38400 corresponds to 0x0005; 57600 corresponds to 0x0006; 115200 corresponds to 0x0007;

For example: Write changes address baud rate 9600 to 38400 (38400 corresponds to 0x0005)

Host request: send data as 01 06 00 03 00 05 B9 C9							
01	06	00	03	00	05	B9	C9
Device address	function code	Preset register address high byte	Preset register address lower byte	Place data high byte	Place data low byte	CRCL	CRCH
Slave reply: Return data is 01 06 00 03 00 05 B9 C9							
01	06	00	03	00	05	B9	C9
Device address	function code	Preset register address high byte	Preset register address lower byte	Place data high byte	Place data low byte	CRCL	CRCH

5. Set the output mode of the sensor. After the output mode is modified, the module needs to be powered on again to take effect. At present, it supports change output mode, continuous output mode and response output mode. The default mode of the module is change output mode;

the change output mode corresponds to 0x0001; the continuous output mode corresponds to 0x0002; the response output mode corresponds to 0x0003;

For example: change the output mode of sensor with device address 1 to continuous output mode (continuous output mode corresponds to 0x0002)

Host request: send data as 01 06 00 04 00 02 49 CA							
01	06	00	04		00	02	49 CA
Device addres	function code	Preset register address high byte	Preset register address lower byte	Place data high byte	Place data low byte	CRCL	CRCH
Slave reply: Return data is 01 06 00 04 00 02 49 CA							
01	06	00	04		00	02	49 CA
Device addres	function code	Preset register address high byte	Preset register address lower byte	Place data high byte	Place data low byte	CRCL	CRCH

6. Set the output frequency of the sensor data in the continuous output mode. After the output frequency is modified, the module needs to be powered on again to take effect. At present, it supports 10Hz, 25Hz, 50Hz, 100Hz, and the default data output frequency of the module is 50Hz; 10Hz corresponds to 0x0001; 25Hz corresponds to 0x0002; 50Hz corresponds to 0x0003; 100Hz corresponds to 0x0004;

For example: change the output frequency of the sensor with device address 1 to 100Hz (100Hz corresponds to 0x0004)

Host request: send data as 01 06 00 05 00 04 98 08							
01	06	00	05	00	04	98	08
Device addres	function code	Preset register address high byte	Preset register address lower byte	Place data high byte	Place data low byte	CRCL	CRCH
Slave reply: Return data is 01 06 00 05 00 04 98 08							

01	06	00	05	00	04	98	08
Device addres	function code	Preset register address high byte	Preset register address lower byte	Place data high byte	Place data low byte	CRCL	CRCH

7. If you forget to set the address, you can use the unconditional address query command, which is a special command. When using, the module must be connected to the computer separately.

Get the address information of the sensor

Frame formats: sending

Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	CRCL	CRCH
FF	03	00	02	00	01	30	14

return

Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	CRCL	CRCH
FF	03	02	00	ADDR	*	*

ADDR indicates the address information of the sensor, * indicates the returned value

=====END=====

Ordering information:

1. This sensor is divided into different interface output and communication protocol output. When ordering, please select according to the model specification comparison table. This model is applicable to the model: CCF-NS16-2M、CCF-NS16-4M;
2. Use strictly in accordance with the instructions when using, so as to cause unnecessary losses;
3. Please cut off the power supply during fixed installation;
4. If you find that there is no output of the sensing signal, please observe whether the signal lamp is normally lit. If the normal lighting of the inspection line is correct and whether it is connected, the above problems cannot be solved. Please contact our technical personnel for communication and solution.
5. This sensor is warranted for one year and man-made damage is not covered by the warranty;